



User Guide

First-Principles Estimating, and How to Use This Software

This guide has two parts. Part One explains first-principles estimating itself — the considerations a proper estimator works through before a number ever gets typed in. Part Two walks through Ground Truth Estimator step by step, in the order you'll actually use it, from setting up your workspace and measuring quantities off your drawings to sending out a finished estimate.

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First-Principles Estimating

“First-principles” estimating means building a price up from its real components — labour hours, plant hours, and material quantities — rather than pulling a single per-metre or per-square-metre figure from memory or from a past job that may not be comparable. It’s slower than a gut-feel number, but it produces a price you can actually defend, line by line, if a client, a quantity surveyor, or your own finance team asks “where did this number come from?”

“A single lump-sum rate is a guess wearing a suit. A build-up is a guess you can cross-examine.”

The ten considerations behind every sound estimate

These apply whatever software you use — they’re the discipline itself, not a feature list. Ground Truth Estimator is built to walk you through them in this order.

1. Understand the scope and the site

Before any number is typed, read the drawings, specification, and site conditions. Access, ground conditions, staging area, existing services, and site constraints all change what labour and plant an item actually needs — the same trench costs differently in open paddock versus a live urban road.

2. Break the work into logical categories

Group the project into work packages that make sense for how it will actually be built and managed — typically earthworks and site works, roads and pavements, drainage and pipelines, and structures. This keeps the estimate organised and makes it possible to compare category totals against past jobs.

3. Quantify each item properly (quantity take-off)

Measure real quantities off the drawings and specification — lengths, areas, volumes, counts. An estimate is only as good as its quantities; a first-principles rate applied to a wrong quantity still produces a wrong price. Ground Truth Estimator’s Takeoff tab lets you measure lengths, areas, and counts straight off a PDF drawing and send them into the estimate — see Step 4 in Part Two.

4. Build the rate up from labour, plant, and material

For every line item, work out what labour it actually takes (whose time, how many hours), what plant it needs (machine and hire hours), and what material goes into it (quantity and unit cost) — then add those three together to get the unit rate. This is the core of “first principles.”

5. Keep units of measure consistent

A rate quoted per linear metre is meaningless multiplied against a quantity measured in square metres. Confirm every item’s unit matches how it was actually quantified, and be explicit when converting between metric and imperial so nothing gets lost in translation.

6. Assess risk as its own exercise — not a blanket percentage

Rather than adding a flat “10% for contingency” across the board, list out the specific things that could go wrong — weather delays, unknown ground conditions, market price movement, programme slippage,

safety incidents — and estimate a probability and a cost impact for each. The sum of those gives a contingency figure you can actually explain, item by item, rather than a number picked to feel safe.

7. Weigh location- and season-specific risk

Where and when the work happens matters. A project programmed through a wet season or a cyclone-prone period carries real, quantifiable weather risk that a generic contingency percentage won't capture.

8. Apply markups deliberately, and keep tax separate

Preliminaries / Indirect Job Costs (site establishment, supervision, temporary works), overhead (your business's running costs), and margin (profit) are three different things and should be considered separately, not lumped into one “mark-up.” Sales tax (GST, VAT, or your local equivalent) is not a cost to the business at all — it should be shown separately, not folded into the price you're trying to win the job on.

Where you have enough information to do it properly, apply this same discipline to preliminaries itself: build it up from the real components — site supervision, temporary services, site facilities, insurances, bonds — rather than estimating a flat percentage of direct cost. See Step 7 in Part Two.

9. Sanity-check the total

Once the build-up is complete, step back and compare the total — and the category breakdown — against past projects, industry benchmarks, or a rough top-down check. A first-principles estimate that comes out wildly different from experience is a signal to go back and check assumptions, not necessarily proof the number is right.

10. Document assumptions and keep rates current

Rates go stale. Labour agreements change, plant hire costs move, material prices fluctuate. A rate library is only trustworthy if it's actively maintained against real supplier quotes and current agreements — treat every stored rate as a placeholder until you've checked it recently.

Using Ground Truth Estimator, Step by Step

The software mirrors the discipline in Part One. Used in this order, it walks you through the same considerations a first-principles estimator works through by hand — just faster, and with every figure stored, reusable, and traceable.

1 Set up your workspace

When you create an account, a workspace is created for your company. Everything you build — rates, projects, risk registers — belongs to that workspace and is never visible to anyone outside your organisation.

Before doing anything else, go to Settings and set your workspace's currency and preferred unit of measure (metric or imperial). Every dollar figure in the app formats to that currency, and any rate or quantity entered in the other unit system will show a small converted equivalent alongside it — without changing what you actually typed.

Where: Left rail → Settings

2 Build your Rate Library first

Before starting a project, populate the Rate Library with your own real labour, plant, material, and subcontract rates — not the indicative placeholders the app starts with. This is the foundation everything else is built from, so it's worth getting right before you start pricing actual work.

Add rows under Labour, Plant & Equipment, Material, and Subcontract. Each rate needs a description, a unit, and a cost per that unit. If you're unsure of a current market rate, the **“Search the web”** button next to each row opens a web search for that item so you can check current supplier, hire, or subcontractor pricing — the search results are for your own reference; nothing is applied automatically.

Where: Left rail → Rate Library

3 Start a project

From Projects, create a new project — either blank, or from the example project if you want to see a fully worked estimate first. Give it a name, client, location, and prepared-by details; these appear on exports.

A new project comes pre-loaded with four standard categories (Earthworks & Site Works, Roads & Pavements, Drainage & Pipelines, Structures), which you can rename, reorder, or add to as the project requires.

Where: Left rail → Projects → New Project

4 Take off quantities from drawings

The Takeoff tab is where quantity take-off (consideration 3 in Part One) happens. Upload a 2D drawing as a PDF, tell the app its scale, then trace lengths, areas, and counts directly on the drawing — and send each measured quantity straight into the Bill of Quantities as a line item, ready for its rate to be built up in Step 5.

Upload a drawing

Click **+ Upload drawing** and choose a PDF. Each drawing appears as a button along the top of the tab — click one to open it. For a multi-page drawing set, use **← Page** and **Page →** to move between sheets. Drawings are stored privately in your workspace, alongside the project.

Set the scale — once for each page

Measurements are only meaningful once the app knows the drawing's scale, so this comes first. Click **Set scale**, then click two points on the drawing a known distance apart — the ends of the scale bar in the title block are ideal, or a dimensioned feature. A small box then asks for the real-world distance between those two points and its unit (for example **100** and **m**); click **Set scale** to save it. The blue information bar confirms “Page 1 scale set: calibrated in m.”

Every page has its own scale, because the sheets in a drawing set are often drawn at different scales. When you open a page that hasn't been calibrated yet and another page of the same drawing has been, the app asks whether this page is drawn at the same scale — click **Use page N's scale** to reuse it in one click, or **Set scale for this page** to calibrate it separately.

Tip — accuracy. Calibrate over the longest known distance you can, zoomed in so you can click precisely: a small miss on a short distance becomes a large error across a whole site. A drawing's printed ratio (for example 1:2500) is usually quoted for its original sheet size, so a reduced PDF won't match it — the scale bar shrinks with the drawing and is always right. Check the result by measuring something whose size you already know.

Zoom in for precise clicking

Use the **–** and **+** buttons in the toolbar, or hold **Ctrl** and roll the mouse wheel over the drawing (a trackpad pinch works too). Wheel zoom centres on your mouse pointer, so you can point at a detail and roll straight in to it; scroll to move around a zoomed drawing, and click the percentage to fit it back to the width. Zoom only changes how large the drawing is shown — measurements and scales are equally accurate at any zoom level.

Measure lengths, areas, and counts

- **+ Length** — click points along a line (a kerb, a fence, a pipe run); for a bend, click more points. The total is the sum of the straight segments between your clicks.
- **+ Area** — click each corner of a shape (a pad, a car park, a laydown area). The shape closes automatically, and any outline works, not just rectangles.
- **+ Count** — click each item to count (pits, light poles, trees). Counting doesn't need a scale.

While you click, the information bar shows a live running total — for example “Length so far: 124.6 m” or “Area so far: 2,310 m²” — so you can check a measurement before saving it. Click **Finish**, give the measurement an optional label, and click **Save measurement**. **Cancel** discards the

measurement in progress, and moving to another page does too, so points can never land on the wrong sheet. Length and area need the current page's scale; if it isn't set yet, the app tells you before anything is saved.

Send a quantity to the estimate

Saved measurements are listed beside the drawing for the page you're on, each with its quantity and unit (m, m², or each). Click → **Send to estimate** on one, pick the category, adjust the description or quantity if you need to, and click **Add line item**. A new line item appears on the Estimate tab with that quantity and unit, ready for its rate to be built up; the measurement is then marked “✓ In estimate” and drawn in green on the drawing (measurements not yet sent are blue).

A saved measurement keeps the value it was measured with — setting a page's scale again doesn't recalculate measurements already taken, so delete and re-measure any taken with the old scale. Deleting a measurement or a whole drawing never removes line items already sent to the estimate; deleting that line item from the estimate simply returns the measurement to “not yet sent.”

Where: Project → Estimate group → Takeoff tab

5 Build the estimate — and import line items

Inside a project, the Estimate tab is where the first-principles build-up actually happens. Add line items under each category — or send them across from the Takeoff tab (Step 4) — give each a description, unit, and quantity, then build up its rate from your Rate Library, pulling in the specific labour, plant, material, and subcontract components that make up that item, each with its own per-unit consumption. To add a category of your own, click **+ Add category**, type its name, and press **Enter**.

The line item's rate is the sum of those components — exactly the discipline described in Part One. If a component's unit is in the other measurement system from your workspace default, a small converted figure appears alongside it so you can sanity-check without doing the maths by hand.

Flat rate. Not every item needs a full build-up. Open any line item and switch it to “Flat rate” to type its unit rate directly instead — useful for a quick allowance or a subcontractor's lump-sum quote for the whole item. Switching back to “Build up” doesn't lose the flat rate you typed; it's simply not used while build-up mode is selected.

Import line items. The “ Import line items” button lets you bring in priced items from another spreadsheet instead of typing them in one by one. Upload an Excel (.xlsx) or .csv file — choosing which tab to import from if the workbook has more than one — or paste rows copied directly from Excel or Google Sheets. The importer finds the column headings (Description, Unit, Quantity, Rate — however far down the sheet they sit) and reads every row beneath them: a row with a description but no usable quantity or rate becomes a new category heading, and a priced row underneath becomes a line item in that category. A preview with a checkbox per item is shown before anything is actually imported, and each category can be renamed or sent into an existing category. Imported items come in as Flat rate items, since a spreadsheet row typically carries one ready-made rate rather than a labour/plant/material breakdown.

Where: Project → Estimate tab

Work through Risk & Location

Move to the Risk & Location tab and build the project's risk register as its own exercise, separate from the base estimate — add each identified risk with a description, a probability (as a %), and a cost impact if it occurs. The tab totals these into a risk allowance you can see and defend line by line, rather than a single guessed contingency percentage. For a risk where the cost itself is uncertain, not just whether it happens, the optional **Min** and **Max** columns let you enter a 3-point estimate instead of one fixed figure — leave them blank for risks you already know the cost of; where they're set, the Summary tab's risk-adjusted price range treats that risk as a spread rather than a single number.

The cost impact — and Min/Max — are always your own entry. None of the lookups below know this project's dollar exposure, so none of them pre-fill a cost figure; each can only suggest a probability and a description from the hazard or trend it finds. For the market lookup, a reasonable starting point is the value of the affected material or trade package multiplied by the cost trend shown.

Site & investigation risk turns two judgement calls most estimators make by instinct into an explicit part of the register. Set whether the site is **greenfield** or **brownfield**, and for a brownfield site, enter how many separate services were identified in the Dial Before You Dig referral — more services found suggests a higher probability of an unplanned utility strike. Separately, choosing an investigation level — **Minimal** (desktop study only), **Some** (preliminary boreholes or test pits), or **Detailed** (comprehensive investigation) — suggests a probability of hitting latent conditions the investigation didn't catch, naturally higher for a minimal investigation and lower for a detailed one. As with the five lookups below, only a probability and description are suggested; the cost impact is still yours to set before adding either to the register.

Enter the site location once and use it for five live lookups. The **weather risk lookup** pulls historical climate data — average rainfall, rainy days, and cyclone-zone exposure. The **geotechnical / soil risk lookup** checks free government soil survey data (USDA SSURGO in the United States, CSIRO's national Australian Soil Classification in Australia) and can flag reactive clay, poor drainage, or a shallow water table. The **flood risk lookup** checks FEMA's National Flood Hazard Layer for US sites, and the **seismic risk lookup** checks USGS Design Maps (US) and Geoscience Australia's National Seismic Hazard Assessment (AU) — both can flag an elevated hazard level worth a specific allowance. The **market / price escalation lookup** checks free government cost and materials price statistics — the US Bureau of Labor Statistics, the Australian Bureau of Statistics, the UK's Office for National Statistics, and Eurostat for EU member states — and can flag an above-normal 12-month cost trend worth a bigger escalation allowance.

Each lookup that finds something relevant can suggest a pre-filled risk item with a probability, ready to add straight to the register or adjust first. Outside the regions a lookup covers, it will say so plainly rather than guess — add a risk row manually instead. All five are a desktop-level early warning, not a substitute for a real site investigation, current supplier pricing, the actual FEMA FIRM panel, or a project-specific seismic assessment.

Where: Project → Risk & Location tab

7 Review the Summary and export

The Summary tab rolls everything up: direct costs by category, the risk allowance, and your markups — preliminaries / indirect job costs, contingency, overhead, and margin — applied on top. Sales tax (GST / VAT / Sales tax, depending on your workspace) is shown as a separate line, not folded into the price itself, and there's a separate line for client-side administrative cost where relevant.

Preliminaries / Indirect Job Costs can be priced two ways. **Simple %** (the default) applies a flat percentage of direct cost — quick, and reasonable for an early, rough-order estimate. **Itemised build-up** applies the same first-principles discipline from Part One to preliminaries itself: instead of a guessed percentage, list the real components — site supervision, temporary services, site facilities, insurances, bonds and guarantees, mobilisation — each priced as either a one-off **Fixed** cost or a **\$/week Time-related** cost, multiplied automatically by the project duration you enter. Two **Quick add** panels above the itemised list save re-typing common site-staff roles and pay items.

Client Cost can be priced the same two ways. This is the client's (Principal's) own project administration, shown separately after the contract price. **Simple %** (the default) applies a flat percentage of contract price. **Itemised build-up** lists it out instead — client-side roles (Project Director, Design Manager, PUP Coordinator, and others) as **\$/week Time-related** items, and one-off **Fixed** items like a geotechnical investigation, survey, or environmental approvals — with its own project duration field, kept separate from the contractor's construction duration, since the client's own administration usually spans more of the project (concept, design, delivery, finalisation) than just the construction period.

The **Risk-adjusted price range** table simulates thousands of outcomes of your risk register — each risk either happening, at its stated probability, or not, and spreading across its Min/Max range where you've set one — and shows a best-case, expected, and worst-case figure for the risk allowance, contract price, and total project cost. Treat it as a decision tool for setting your own contingency, not something to hand a client on a competitive fixed-price bid — they want your price, not your worst case.

Cash Flow spreads the total project cost evenly across however many months you set, starting from the project's date, shown as a month-by-month table with a simple bar per month — a straight-line spread for a first-pass view of spend over time, not a contractual payment schedule.

Adjust markup percentages here and watch the total update live. When the estimate is ready, export it to CSV for use in a proposal, tender, or your own reporting.

Where: Project → Summary tab

8 Programme — schedule your work

The Programme tab gives each category a planned start and end date, so Earned Value (see Step 10) can compare real progress against a realistic time-phased budget instead of an even spread across the whole job. A category left unscheduled is still included in the total — its budget is simply spread evenly across the whole programme window until you set its dates.

Duration is a plain read-out — the number of days between a category's own start and end — so it's never something you type in directly, only ever a reflection of the two dates.

Predecessor and Lag link one category's start to another's finish, similar to Microsoft Project's Predecessor field (Finish-to-Start only, the most common case). Set “Roads & Pavements” to follow “Earthworks,” for example, and its planned start recalculates automatically as the day after Earthworks finishes, plus whatever Lag (in days) you set — a positive lag adds a gap, a negative one allows overlap. Only the start date is linked this way; you still set each category's own end date directly, so a shift upstream moves a start date without silently stretching or shrinking anyone's duration.

Import from Primavera P6. Upload a .xer export and the app rolls each WBS node's activities up to a single earliest-start / latest-end date, ready to map onto your existing categories (or create new ones). Every import is saved as a dated snapshot, so re-importing next month lets you compare “then” against “now” for every category and see exactly how many days each one has shifted — a simple way to re-benchmark the programme monthly without losing the history.

Planned Value is deliberately different from Cash Flow (Step 7) — Cash Flow spreads the whole billing total (including preliminaries, overhead, margin and tax) evenly for a payment forecast, while Planned Value spreads only the direct-cost budget, phased to when each category is actually scheduled, for comparison against real progress.

Where: Project → Programme tab

9 Positions & Actuals — record real cost

These two tabs record what's actually happening on site against the estimate: Positions is the register behind Indirect Job Cost, and Actuals is where every real Direct and Indirect cost is logged. A separate Earned Value tab (Step 10) turns recorded progress into the figures that compare all this against the plan.

Positions is the Indirect Job Cost register — one row per site or support role. Each position is flagged **Wage** or **Salaried**. A Wage position (hourly site labour, for example) is paid a rate that already IS the true cost, so no multiplier is ever applied — it's locked at 1.00×. A Salaried position (typically an onsite overhead role — Site Engineer, Foreman, Safety Officer) is paid a fixed salary that doesn't cover everything it actually costs, so an editable **Reimbursable Cost Multiplier (RCM)** loads it up into a fully-recovered rate, capturing on-costs like leave, super, insurances, IT access, and site facilities. Each position's rate can be set per hour, per week, or per year, and the tab converts everything to one comparable weekly total.

Actuals has two ledgers, and every entry in both can be edited in place afterwards — not just deleted and re-entered. The **Direct Job Cost ledger** logs every real Labour, Plant, and Material cost by picking a Rate Library item and a quantity — the amount is calculated automatically from the same rates used in the estimate, then frozen at that moment like a recorded invoice or timesheet, so correcting the Rate Library afterwards doesn't silently rewrite past actual cost history. Subcontract costs are still typed directly, since a subcontractor's invoice isn't priced off the Rate Library — its description field doubles as an invoice number to track. The **Indirect Job Cost ledger** logs hours worked against a Position; the dollar cost isn't typed in, it's calculated automatically as hours × that position's current fully-loaded rate, so correcting a position's rate later automatically corrects its past actual cost too.

Where: Project → Positions tab, Project → Actuals tab

10 Earned Value — progress and performance

The Earned Value tab is where you type each line item's % complete as work happens on site — Earned Value (EV) is that line item's full budgeted total × its % complete, the per-line-item granularity this app uses for Earned Value Management.

Together, Planned Value (Step 8, Programme), Earned Value, and Actual Cost (Step 9, Actuals, shown alongside EV here for reference) are the three figures behind Earned Value Management — comparing what you expected to spend by now against what you've actually achieved and actually spent.

Where: Project → Earned Value tab

11 Dashboard — see it all in one place

The Dashboard tab rolls the whole project up into one reporting view: cost by category, cash flow, and cost by resource type (labour, plant, material, subcontract), without having to flip between tabs. A one-time **Major Category mapping** lets you group your individual categories under broader headings — Earthworks, Bridgework, Drainage, and so on — so the top-level chart reads at a glance even on a project with many detailed categories. Any category left unmapped still shows up, grouped under “Unmapped,” so nothing silently disappears from the total.

Sections can be shown or hidden, filtered by category or date range, and the whole thing can be printed or saved as a PDF for a client update or an internal report — a customisable snapshot of the project rather than a fixed report.

Where: Project → Dashboard tab

12 Keep it current

A first-principles estimate is only as good as the rates behind it. Revisit the Rate Library periodically against real supplier quotes, current labour agreements, plant hire pricing, and subcontractor pricing — the “Search the web” button is there specifically to make that check quick.

When a revised drawing arrives, upload it on the Takeoff tab alongside the original, set its scale, and re-measure the items that changed. Measurements stay listed under the drawing and page they were taken from, so it's clear which revision each quantity came off — it helps to include the revision in a measurement's label too.

Re-import your schedule from Primavera P6 periodically too — each import is saved as a dated snapshot, so the Programme tab's Schedule history always shows how the plan has actually moved.

If you ever need to reset your password, use the “Forgot password?” link on the login screen; if you try to create a second account with an email that's already registered, the app will tell you directly rather than leaving you wondering why no email arrived.

QUICK REFERENCE

Where do I...?

I want to...	Go to
Set my workspace's currency or unit system	Settings
Add or edit a labour, plant, material, or subcontract rate	Rate Library
Check a current market rate for something	Rate Library → "Search the web" button
Start a new project	Projects → New Project
Upload a PDF drawing, then measure a length, area, or count off it	Project → Takeoff tab → + Upload drawing, then + Length / + Area / + Count
Set a drawing's scale (each page has its own)	Project → Takeoff tab → Set scale
Zoom in on a drawing	Project → Takeoff tab → - / + buttons, or Ctrl + mouse wheel
Send a measured quantity into the estimate	Project → Takeoff tab → measurement → Send to estimate
Add a category or line item to an estimate	Project → Estimate tab
Build up a line item's rate from labour/plant/material/subcontract	Project → Estimate tab → line item
Type a flat rate instead of building one up	Project → Estimate tab → line item → Flat rate
Import line items from a spreadsheet	Project → Estimate tab → Import line items
Add a risk, or run a site risk lookup	Project → Risk & Location tab
Set site nature (greenfield/brownfield), utility count, or investigation level	Project → Risk & Location tab
Adjust preliminaries / indirect job costs, contingency, overhead, or margin	Project → Summary tab
Switch preliminaries, or client cost, to an itemised build-up	Project → Summary tab → Preliminaries / Client Cost
Quick-add a common site-staff or client-team role	Project → Summary tab → Quick add
See spend spread over time	Project → Summary tab → Cash Flow
Export the estimate	Project → Summary tab → Export CSV
Schedule a category, or import a schedule from Primavera P6	Project → Programme tab
Link one category's start to another's finish (Predecessor)	Project → Programme tab → Predecessor / Lag
Compare this month's schedule against a past import	Project → Programme tab → Schedule history
Set up a Wage or Salaried position and its Reimbursable Cost Multiplier	Project → Positions tab
Log a Labour, Plant, or Material cost against a Rate Library item and quantity	Project → Actuals tab → Direct Job Cost ledger
Log a Subcontract cost with an invoice number	Project → Actuals tab → Direct Job Cost ledger
Log hours worked against a Position	Project → Actuals tab → Indirect Job Cost ledger
Edit an existing actual cost or hours entry	Project → Actuals tab → click into any field
Update a line item's % complete, or see Earned Value	Project → Earned Value tab
See cost by category, cash flow, and resource type in one view	Project → Dashboard tab
Reset a forgotten password	Login screen → Forgot password?
Re-read this guide	Left rail → Help

Your account

Your workspace and everything in it — rates, projects, drawings, risk registers — is private to your organisation. Signing in with an email that already has an account will tell you so directly rather than silently failing; use “Forgot password?” on the login screen at any time to regain access without contacting support.

This guide covers the app as it stands today and will be updated as new features are added. If something here doesn't match what you see on screen, the in-app Help page always reflects the current version.